

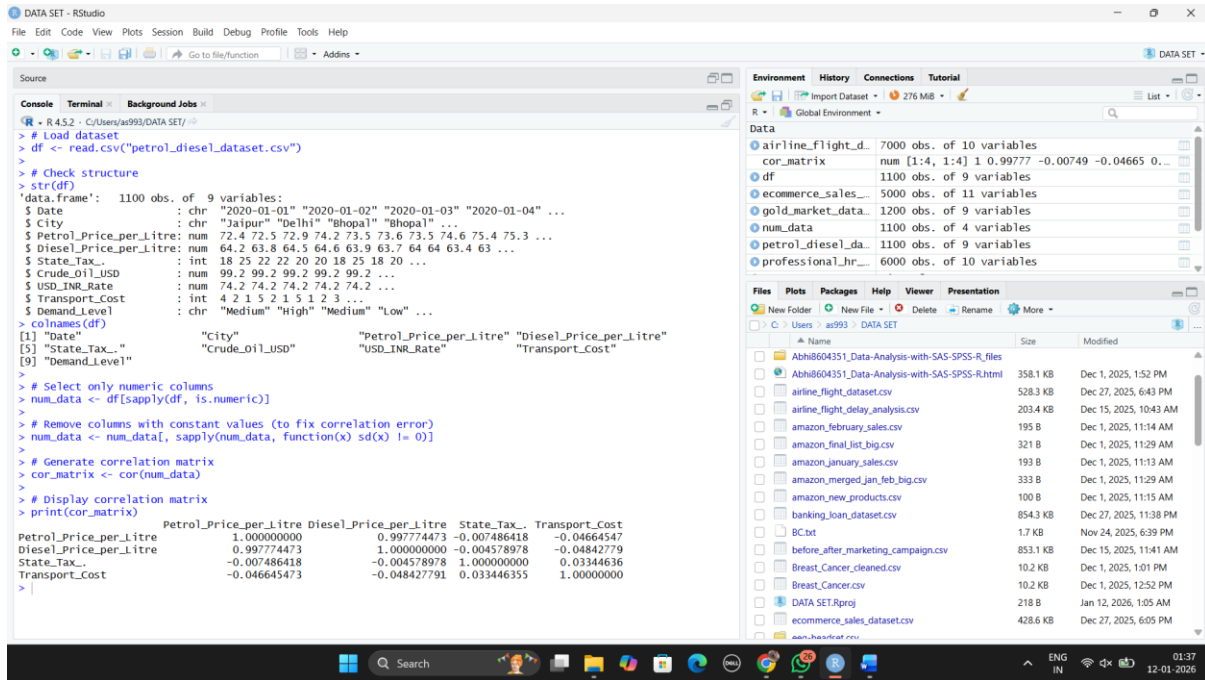
Sheth L.U.J. College of Arts And Sir M.V. College of Science and Commerce

Data Analysis with SAS / SPSS / R

PRACTICAL NO 12 MODULE 2

AIM: Generating correlation matrices using cor() (R).

Output:



The screenshot displays the RStudio interface with the following components:

- Source Panel:** Contains R code for loading a dataset, checking its structure, selecting numeric columns, and generating a correlation matrix.
- Console Panel:** Shows the output of the R code, including the structure of the dataset and the resulting correlation matrix.
- Environment Panel:** Lists the objects in the R environment, including the dataset and the correlation matrix.
- Files Panel:** Shows the file explorer with various CSV files and R scripts.

```
> # Load dataset
> df <- read.csv("petrol_diesel_dataset.csv")
> # Check structure
> str(df)
'data.frame': 1100 obs. of 9 variables:
 $ Date      : chr "2020-01-01" "2020-01-02" "2020-01-03" "2020-01-04" ...
 $ City      : chr "Jaipur" "Delhi" "Bhopal" ...
 $ Petrol_Price_per_Litre: num 72.4 72.5 72.9 74.2 73.5 73.6 73.5 74.6 75.4 75.3 ...
 $ Diesel_Price_per_Litre: num 64.2 63.8 64.5 64.6 63.9 63.7 64 64 63.4 63 ...
 $ State_Tax_ : int 18 25 22 22 20 20 18 25 18 20 ...
 $ Crude_Oil_USD : num 99.2 99.2 99.2 99.2 99.2 ...
 $ USD_INR_Rate  : num 74.2 74.2 74.2 74.2 74.2 ...
 $ Transport_Cost: int 4 2 1 5 2 1 5 1 2 3 ...
 $ Demand_Level : chr "Medium" "High" "Medium" "Low" ...
> colnames(df)
[1] "Date" "City" "Petrol_Price_per_Litre" "Diesel_Price_per_Litre"
[5] "State_Tax_" "Crude_oil_USD" "USD_INR_Rate" "Transport_Cost"
[9] "Demand_Level"
> # Select only numeric columns
> num_data <- df[sapply(df, is.numeric)]
> # Remove columns with constant values (to fix correlation error)
> num_data <- num_data[, sapply(num_data, function(x) sd(x) != 0)]
> # Generate correlation matrix
> cor_matrix <- cor(num_data)
> # Display correlation matrix
> print(cor_matrix)
```

The correlation matrix output is as follows:

	Petrol_Price_per_Litre	Diesel_Price_per_Litre	State_Tax_	Transport_Cost
Petrol_Price_per_Litre	1.000000000	0.997774473	-0.007486418	-0.04664547
Diesel_Price_per_Litre	0.997774473	1.000000000	-0.004578978	-0.04842779
State_Tax_	-0.007486418	-0.004578978	1.000000000	0.03344636
Transport_Cost	-0.046645473	-0.048427791	0.033446355	1.000000000

NAME : ABHISHEK DINESH SINGH
ROLL NO: S116